

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

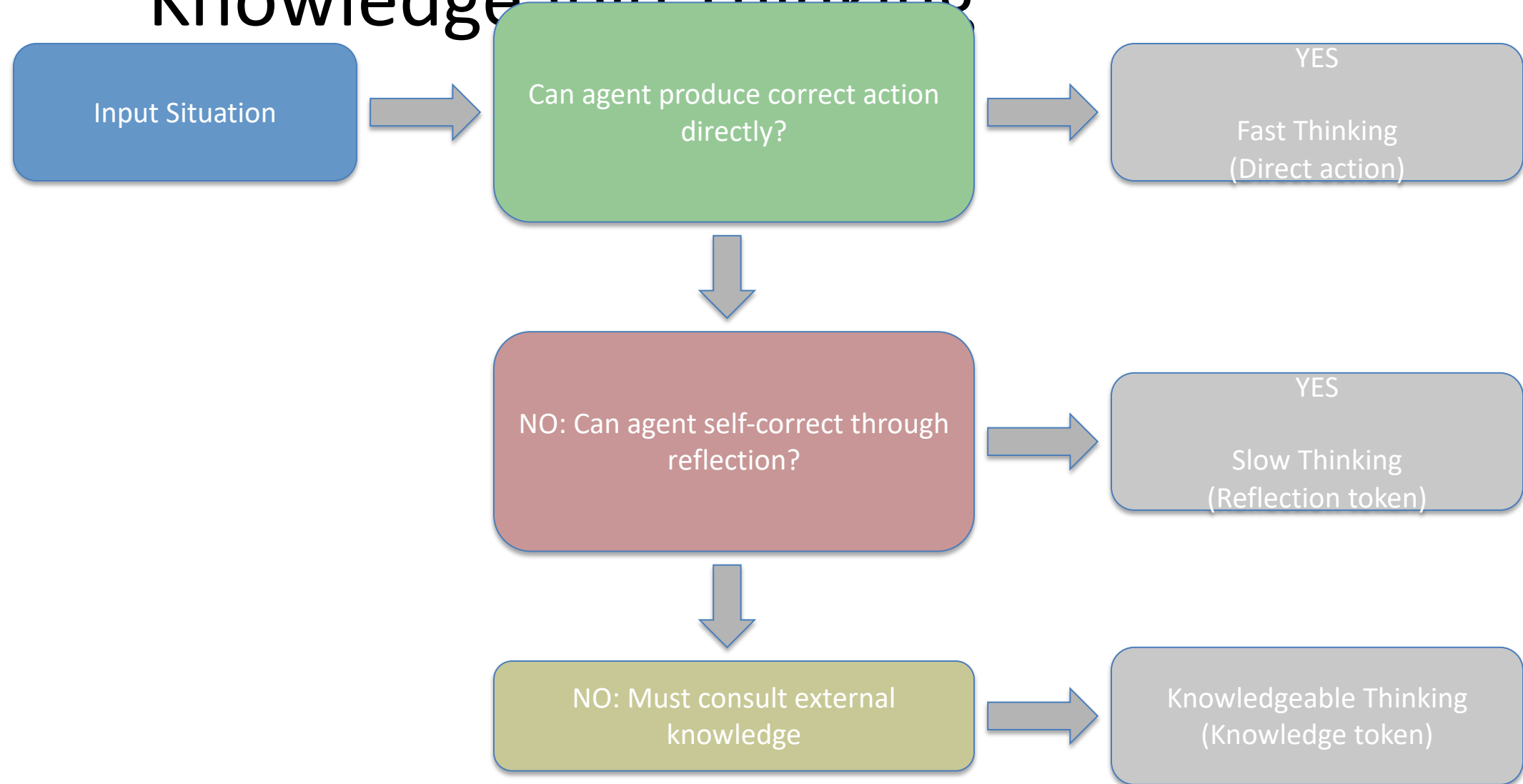
Enable agents to autonomously
regulate knowledge utilization
through situational self-awareness.

Source: arxiv.org/abs/2504.03553

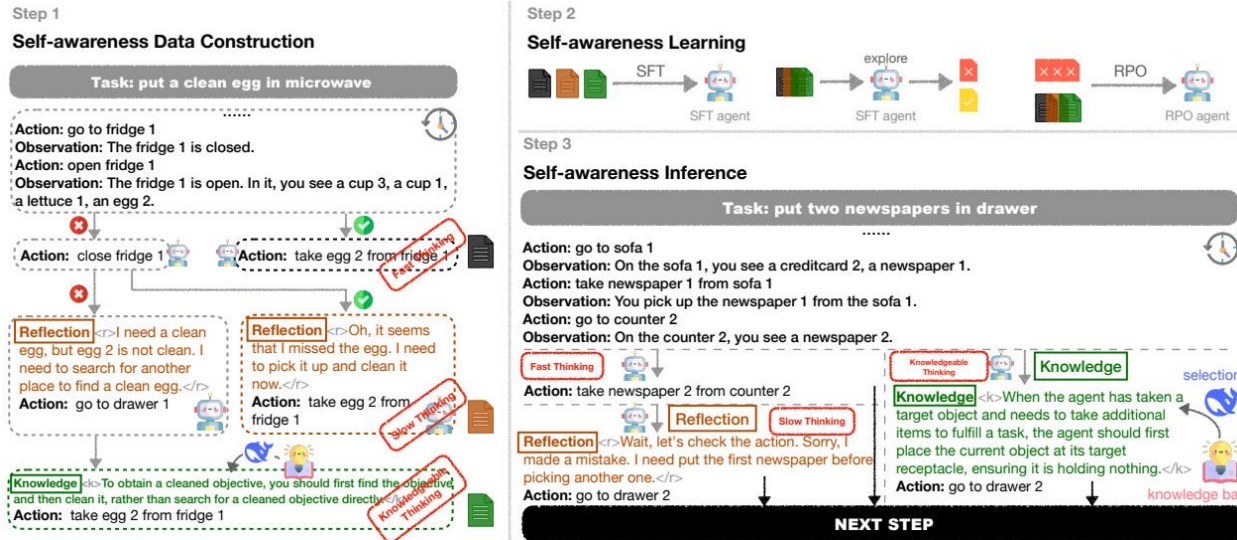
Current Agent Training Uses 'Flood Irrigation'

- Indiscriminate Knowledge Injection Hurts Efficiency.
- Existing paradigms apply one-size-fits-all strategies:
 - - Trajectory Imitation (e.g., ReAct): Blindly copies actions without understanding.
 - - Trial-and-Error (e.g., Reflexion): Overuses reflection even for simple steps.
 - - Knowledge-Augmented (e.g., KnowAgent, WKM): Forces external queries continuously.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction -> Two-Stage Training



nce

Step 1: Self-Awareness Data Construction

Agent self-explores environments. A heuristic criterion classifies the situation (fast, slow, knowledgeable). Special tokens are inserted.

Step 2: Two-Stage Training

First, Supervised Fine-Tuning (SFT) teaches initial self-awareness patterns. Then, Reward-based Policy Optimization (RPO) refines the ability.

Step 3: Self-Aware Inference

During deployment, the agent generates special tokens that trigger either direct action, reflection, or knowledge retrieval from an external KB.

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where

(a) Task: put a hot mug in cabinet	
Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <<>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</> Action: take egg 1 from microwave 1 Observation: Nothing happens.	OpenAI-O1
Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <<>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</> Action: heat mug 1 with microwave 1	KnowSelf (Llama-8B)
(b) Task: put a saltshaker in drawer	
Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <<>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	DeepSeek-R1
Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <<>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</> Action: put saltshaker 1 in/on drawer 1	KnowSelf (Llama-8B)

Task 1: Put a hot mug in cabinet

- OpenAI-O1: Incorrectly tries to take an egg from the microwave after opening it.
- KnowSelf: Uses a Reflection token to realize it should heat the mug using the open microwave.

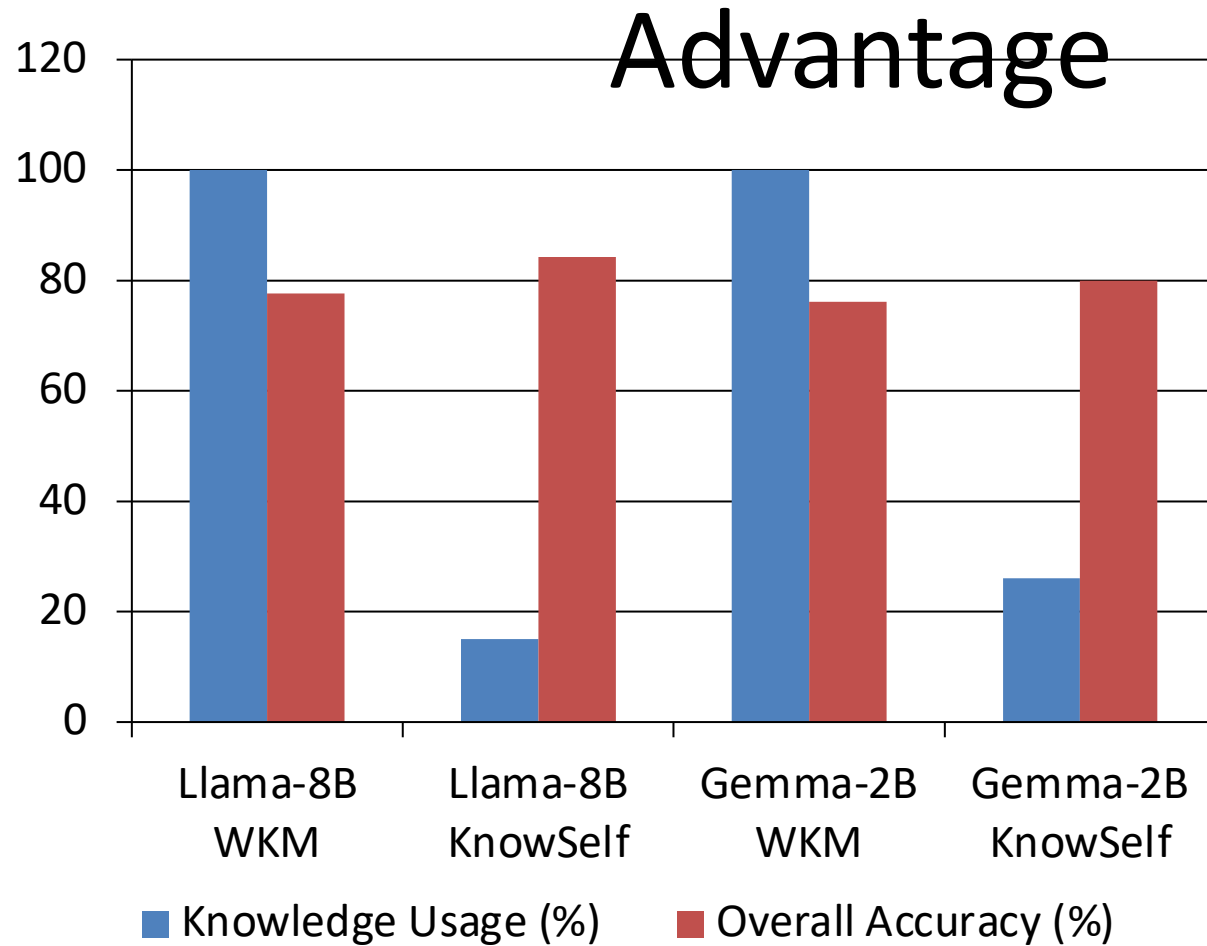
Task 2: Put a saltshaker in drawer

- DeepSeek-R1: Reflects that the drawer is open and tries to close it first (fails).
- KnowSelf: Applies a Knowledge token noting that the agent should place the object without removing unrelated items.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.3	74.2	69.6	85.7	77.8	64.7	76.1
GPT-4o	Reflexion	0%	100	87.1	73.9	90.5	83.3	70.6	85.1
GPT-4o	ExpeL	100%	95.8	83.9	69.6	81.0	88.9	52.9	79.9
Llama-8B	ReAct	0%	33.3	3.2	0.0	57.1	66.7	23.5	27.6
Llama-8B	Reflexion	0%	63.0	22.7	5.9	64.3	86.4	50.0	51.5
Llama-8B	ExpeL	100%	83.3	32.3	30.4	23.8	55.6	17.6	41.0
Llama-8B	ETO	0%	91.7	70.6	82.6	61.9	88.9	64.7	78.4
Llama-8B	KnowAgent	100%	87.5	93.6	65.2	66.7	61.1	64.7	75.4
Llama-8B	WKM	100%	95.8	87.1	87.0	61.9	66.7	52.9	77.6
Llama-8B	KnowSelf	15.0%	91.7	87.1	91.3	85.7	77.8	64.7	84.3
Gemma-2B	ReAct	0%	0.0	9.7	0.0	4.8	44.4	0.0	9.0
Gemma-2B	ETO	0%	91.7	83.9	78.3	52.4	77.8	29.4	71.6
Gemma-2B	KnowAgent	100%	91.7	90.3	69.6	71.4	66.7	41.2	73.9
Gemma-2B	WKM	100%	91.7	87.1	78.3	71.4	61.1	52.9	76.1
Gemma-2B	KnowSelf	26.4%	87.5	93.6	73.9	76.2	83.3	52.9	79.9

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



Efficiency Advantage over Full-Knowledge Methods

- Minimal Knowledge: KnowSelf uses only ~15% (Llama-8B) and 26% (Gemma-2B) knowledge queries compared to 100% for KnowAgent/WKM/ExpeL.
- Maximum Impact: Despite the minimal queries, it achieves the best overall scores (84.33% and 79.85%).
- Scaling Behavior: The approach scales down effectively. Even the smaller Gemma-2B model reaches 79.85% with KnowSelf, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- 1. Indiscriminate knowledge injection is wasteful and suboptimal.
- 2. Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- 3. KnowSelf with Llama-8B outperforms strong baselines including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).
- 4. Only 15-26% knowledge usage needed versus 100% for competing methods.